DEVADATHAN M

Mechanical Design & CAD Engineer

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PROFESSIONAL SUMMARY

Mechanical engineer with 2+ years of experience across CAD design, prototype fabrication, and technical training, now applying that engineering background to operations coordination. Proficient in SOLIDWORKS, CATIA, AutoCAD, and Fusion 360, with hands-on experience carrying designs from concept through fabrication and testing. Currently pursuing an NPTEL certification in Operations and Supply Chain Management to formalize a growing focus on process and resource coordination.

CORE COMPETENCIES

Design & CAD	R&D & Prototyping	Operations	Training
3D Parametric Modeling	Prototype Fabrication	Resource Scheduling	CAD Instruction
Assembly Design	Design Iteration	Process Mapping	Project Mentoring
GD&T	Functional Testing	Documentation	3D Printing Support
Engineering Drawings	Tolerance & Fit	Cross-Functional Coordination	Technical Workshops
DFM			

PROFESSIONAL EXPERIENCE

Operations Coordinator | IIEMS (Indian Institute of Emergency Medical Services)

Kottayam, Kerala — *July 2026 – Present*

- Coordinate daily departmental operations, resource scheduling, and documentation for emergency medical training programs, liaising between administration, instructors, and participants.
- Manage batch rosters, facility readiness, and equipment requirements, resolving scheduling bottlenecks to keep programs on track.
- Apply Operations and Supply Chain Management coursework (NPTEL) to improve workflow planning and resource utilization.

Mechanical Faculty & CAD Trainer | CADD Centre Kottayam

Kottayam, Kerala — *June 2025 – July 2026*

- Delivered hands-on CAD training in SOLIDWORKS, CATIA, and AutoCAD, covering 2D drafting through 3D parametric modeling and assembly design.
- Mentored student design projects from concept through CAD implementation, including design validation and fabrication feasibility.
- Trained learners in GD&T and manufacturing-oriented design practices, and supported troubleshooting of modeling and assembly errors.

Mechanical Engineer – R&D | Agrowtein

Kerala — *August 2024 – December 2025 (full-time to June 2025; project-based thereafter)*

- Developed and iterated parametric CAD models for mechanical components and mechanisms in automation-focused agritech equipment.
- Drove design revisions based on prototype testing and fabrication feedback, applying tolerance, fit, and manufacturability considerations.
- Worked directly with fabricators through prototype build, assembly, and troubleshooting, diagnosing structural and functional issues from field testing.

ENGINEERING PROJECTS

Cardamom Cleaning Machine

- Ran a Design of Experiments (DOE) study to identify key parameters affecting cleaning quality, then designed a mechanism to remove debris while minimizing seed damage.
- Took the design from CAD through a working prototype, correcting fit and manufacturability issues found during fabrication and testing.

Semi-Automatic Steaming & Folding Machine

- Designed a semi-automatic mechanism to streamline garment steaming and folding, from concept through a fabricated working prototype.

NASA Space Apps Challenge

- Contributed system-workflow design for a forest-fire monitoring and rapid-response concept.

COMPETITIONS & TECHNICAL ENGAGEMENTS

- 11th Go-Kart Design Challenge (national-level): contributed to chassis structure, steering geometry, and braking-system design, plus fabrication and testing.
- SEVC Stellar (national-level EV competition): contributed to EV chassis integration and mechanical design reviews.
- Smart India Hackathon 2024: volunteered on participant coordination and event logistics for the college-hosted event.
- Provided 3D printing support to student teams — print setup, slicing, and design-for-printability guidance — to help convert CAD models into working prototypes.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Operations and Supply Chain Management — NPTEL (in progress): process flow analysis, supply chain logistics, inventory management, demand forecasting.

EDUCATION

- **B.Tech, Mechanical Engineering** — Amal Jyothi College of Engineering and Technology (Autonomous), 2022–2025
- **Diploma, Mechanical Engineering** — Government Polytechnic College Kottayam, 2020–2022 (First Class with Distinction)
- **Higher Secondary (Computer Science)** — J.J. Murphy Memorial HSS, Yendayar, 2020–2022, 82%
- **High School** — Government Technical High School, Kanjirappally, 2020, Full A+

INDUSTRIAL TRAINING

- **Cochin Shipyard Ltd. —** Marine & heavy engineering internship covering ship construction, large-scale fabrication, and dockyard operations.
- **Kerala Chandra Pvt. Ltd. —** Manufacturing & maintenance training in preventive/corrective maintenance, equipment inspection, and shop-floor safety practices.